Intelligent Multi-Label Triage for Hematological Comorbidities using BERT

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***Abstract*—Modern triage requires simultaneous evaluation of multiple organ systems; single-label classifiers fail to capture comorbidity. We address clinical routing from basic lab parameters and a minimal adaptive interview triggered when a value is out of reference range. We posit that a BERT-based model pre-trained in the medical domain (Masked Language Modelling, MLM) can route patients to eight classes (primary care, specialists, emergency SOR). We use a hybrid data approach: MIMIC-III/IV (specialists and SOR cases) and Synthea (chronic trajectories and primary care). Labels come from ICD-to-specialist mapping; lab results are quantised for triage. We focus on methodology: MLM pre-training on the hybrid corpus, multi-label fine-tuning with a clinically cost-sensitive loss, and "honest" patient-level validation. Results include ROC curves, calibration (ECE), Precision-Recall curves, confusion matrices, and attention visualisation for interpretability. High ROC AUC for SOR and low ECE support human-in-the-loop deployment with emphasis on safety.**

I. Introduction

Demographic ageing has led to a sharp rise in patients with multiple chronic conditions. In acute and primary care, triage and referral decisions must often be made on the basis of limited history and a small set of tests-for example basic blood count and biochemistry-before a full workup is available. At the same time, missing an urgent case (e.g. severe anaemia or acute kidney injury) or failing to flag the need for several specialists at once can delay treatment and worsen outcomes. We therefore prioritise **safety**: any decision-support system should assist, not replace, the physician, and must correctly identify cases requiring emergency care (SOR) or multi-specialist input (e.g. simultaneous haematology, nephrology and cardiology involvement).

Traditional triage tools rely on rule-based logic or single-label classifiers. Rule-based systems become hard to maintain when symptom descriptions are ambiguous or when the volume of free-text notes grows; they also tend to use fixed thresholds on lab values, which ignore patient-specific baselines and trends. Single-label models, which assign each case to exactly one class (e.g. one specialist), are structurally unsuited to comorbidity: increasing the probability of a cardiology referral forces a decrease in the probability of a pulmonology referral, even when both are clinically indicated. In reality, the triage function should output a **multi-label** vector-independent probabilities for several specialist queues-so that multiple referrals can be activated when the clinical picture warrants it.

In this article we posit that a **BERT-based model pre-trained in the medical domain using Masked Language Modelling (MLM) can appropriately route patients** to multiple classes at once, with interpretability provided by attention visualisation. Our objective is to propose and validate a pipeline that combines (1) quantisation of laboratory parameters into semantic tokens, (2) a minimal **adaptive interview** triggered when a value falls outside the reference range, and (3) multi-label BERT fine-tuned with a loss that reflects clinical costs (e.g. higher cost for missing SOR than for a false alarm). We define the "study" that triggers the interview in a reproducible way (parameter, reference range, question from a bank, answer token), so that the flow from lab result to routing suggestion is deterministic and auditable. We **focus solely on the methodology and results of this model**, without comparison to other classifiers.

Clinical text classification and triage have a long history. Early systems relied on rule-based keyword extraction and hand-crafted thresholds; later, support vector machines (SVMs) and logistic regression were applied to bag-of-words or simple feature representations. These approaches often ignored negation, temporal order, and the interplay between structured data (e.g. lab values) and free text, which limits their reliability in safety-critical settings. With the rise of deep learning, recurrent neural networks (RNNs), in particular Long Short-Term Memory (LSTM) networks, became the standard for modelling clinical sequences [1]. They can capture local dependencies but suffer from long-range dependency issues: information from the beginning of a long note may be poorly propagated to later tokens. The Transformer architecture [2] replaced recurrence with self-attention, processing the full sequence in parallel and allowing direct dependencies between any pair of positions. Bidirectional Encoder Representations from Transformers (BERT) [3] pre-train such encoders on large text corpora with objectives such as Masked Language Modelling (MLM), then fine-tune them for downstream tasks. MLM consists of masking a fraction of input tokens and training the model to predict them from context; this encourages rich contextual representations that transfer well to classification and other tasks. In the biomedical domain, variants such as BioBERT and ClinicalBERT have been used mainly for named entity recognition (NER) and single-label phenotyping or classification.

Relation to prior work. Most clinical NLP work using BERT or similar encoders focuses on single-label tasks (e.g. one diagnosis per note) or on NER and phenotyping rather than triage. Rule-based triage systems are still common in practice but require manual maintenance and do not scale to complex, multi-specialist decisions. Our pipeline differs in three ways: (i) multi-label output with sigmoid (no single-class constraint), (ii) in-domain MLM pre-training on the same hybrid corpus

used for fine-tuning, and (iii) an adaptive interview driven by reference ranges and a question bank, so that the system can request a minimal set of follow-up answers before routing. We do not compare to other classifiers in this paper; the focus is on methodology and validation of this design.

To our knowledge, few works address **multi-label comorbidity triage** with a BERT-style model trained on a **hybrid corpus** that mixes synthetic longitudinal data (e.g. Synthea) and real critical-care data (e.g. MIMIC-III/IV), and that is coupled with an **adaptive interview** defined by reference ranges (`lab_norms`) and a question bank (`questions.json`). Our contribution is to combine in-domain MLM pre-training, multi-label fine-tuning with a clinically cost-sensitive loss, and strict patient-level validation, and to report discriminatory performance (ROC, Precision-Recall), calibration (Expected Calibration Error, ECE), confusion analysis, and attention-based interpretability.

The remainder of the paper is organised as follows. Section II describes the hybrid corpus and the adaptive interview (data and deployment context). Section III presents the model architecture, pre-training and fine-tuning, and the loss function. Section IV reports experiments and results, including ROC and Precision-Recall curves, confusion matrices, calibration, and attention-based interpretability. Section V summarises findings, limitations, and future work.

II. Data and Methodology

The corpus is a hybrid dataset we constructed specifically for training and evaluating the triage model. It merges two complementary sources. **Synthea** [5] is an open-source synthetic patient generator that simulates longitudinal electronic health records with chronic conditions (e.g. anaemia, diabetes, chronic kidney disease). It provides controlled variety, well-defined labels, and a large number of trajectories at low cost, but its patterns may be more regular than in real clinical practice. **MIMIC-III** (or MIMIC-IV) [4] is a widely used, de-identified critical care database hosted on PhysioNet; it contains real ICU data, including acute presentations (e.g. sepsis, bleeding, acute kidney injury) and narrative clinical notes. Combining both sources allows the model to see both chronic, multi-morbid trajectories and acute, life-threatening events, improving robustness and reducing the risk of overfitting to a single data type.

Records from Synthea and MIMIC were preprocessed into a single token format. Structured laboratory values (e.g. haemoglobin, creatinine, platelets) were **quantised** into semantic tokens based on reference ranges (e.g. `HGB_LOW`, `CREATININE_HIGH`, `PLT_CRITICAL_LOW`), so that the model can learn non-linear risk thresholds rather than raw numbers. Free-text symptoms and diagnoses were mapped to symptom or diagnosis tokens (e.g. `SYMPTOM_CHEST_PAIN_YES`). Each training or validation sample is a sequence of such tokens (with optional special tokens `[CLS]` and `[SEP]`) plus a **multi-label** binary vector of length eight, indicating which specialist referrals apply: primary care (POZ), Gastroenterology, Haematology, Nephrology, Emergency (SOR), Cardiology, Pulmonology, Hepatology. Labels are derived from mapping ICD (International Classification of Diseases) codes to these eight classes via a fixed mapping table (e.g. anaemia-related ICD codes $\rightarrow$ Haematology, acute kidney injury $\rightarrow$ Nephrology or SOR); a single patient can have multiple labels (e.g. both Haematology and Nephrology) when multiple conditions are recorded. The corpus was split into training and validation sets with **strict patient-level separation**: every encounter of a given patient belongs to one set only, so that the model never sees the same patient in both training and validation. This prevents leakage and yields metrics that reflect generalisation to unseen patients. Class distribution is imbalanced: Haematology and Nephrology specialist labels are among the most frequent, while Emergency (SOR) is rarer; the cost-sensitive loss and per-class thresholds are designed to handle this. Table I summarises the dataset characteristics.

Example. A single training sample is a space-separated token sequence followed by one or more label tokens. For instance, `AGE_60 SEX_M HGB_Q5 HCT_Q5 PLT_Q6 MCV_Q5 WBC_Q5 CREATININE_Q9 ALT_Q5` encodes a male aged 60 with lab values in quintiles 5-6 and creatinine in quintile 9 (high). The corresponding multi-label vector has value 1 for Nephrology and 0 for other classes when that encounter is labelled for nephrology referral only. Sequences are truncated or padded to 128 tokens; the model sees `[CLS]` $\oplus$ token sequence $\oplus$ `[SEP]`.

Table I
Dataset characteristics (hybrid corpus: Synthea + MIMIC).

Item	Value
Training samples (encounters)	260 663
Validation samples (encounters)	65 166
Vocabulary size (tokens)	143
Classes	8
Patient-level split	80% / 20%

A. Adaptive interview

In deployment, the system may have only a subset of lab results and no detailed history. When a value falls outside the reference range (e.g. low haemoglobin), it is useful to ask a follow-up question (e.g. "Any recent bleeding?") before making a routing suggestion. We call this an **adaptive interview**: it is **triggered** only when a laboratory value is out of range, and the question is chosen from a bank according to the parameter and the direction of the deviation. Reference ranges are defined in `lab_norms.json` and can depend on age and sex. Quantisation maps each lab value to a discrete token: for example, haemoglobin is split into quintiles or critical bands (e.g. `HGB_Q1` for very low, `HGB_Q5` for normal, `HGB_CRITICAL_LOW` when below a critical threshold). Creatinine similarly becomes `CREATININE_Q7` or `CREATININE_Q9` (high) or `CREATININE_Q10_CRITICAL`. This allows the model to

learn non-linear risk boundaries (e.g. "high creatinine" vs "critical") without regressing on raw numbers. This design keeps the interview minimal (only relevant questions are asked) and makes the link between lab values and questions explicit for safety and audit.

For the pipeline to be reproducible and auditable, we define a **study** as the minimal unit that captures one such interview step. A study includes: (1) the **triggering parameter and its quantised token** (e.g. HGB, token `HGB_LOW`); (2) the **applied reference range** (e.g. low, high, critical_low, critical_high for the relevant demographic group); (3) the **question from the bank** (`questions.json` stores trigger, intent, question text, and optionally SNOMED code); (4) the **answer encoded as a token** (`token_yes` or `token_no`) that is appended to the BERT input sequence before the next prediction. This ensures that the path from lab result → question → answer → routing suggestion is deterministic and can be reviewed for safety and fairness. The model architecture described next consumes these token sequences and outputs the multi-label routing probabilities.

B. Model architecture

A high-level view of the pipeline is as follows. The input is a sequence of tokens (demographics, quantised lab values, composite features, symptoms, and optional interview answers). These tokens are embedded and passed through a BERT encoder that was first pre-trained on the same corpus with Masked Language Modelling (MLM), then fine-tuned for multi-label classification. The representation of the special `[CLS]` token is fed into a linear layer with eight outputs, each passed through a **sigmoid** (not softmax) so that the model outputs eight independent probabilities in $[0, 1]$. Figure 1 illustrates this architecture.

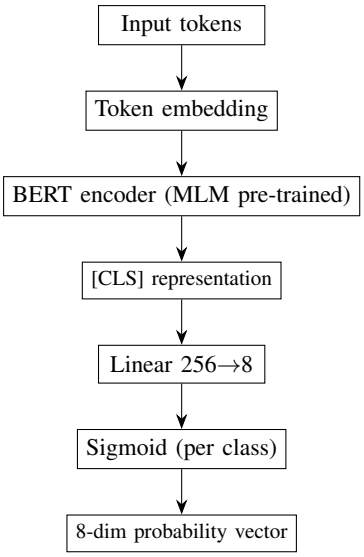

Figure 1. Model architecture: input token sequence → embedding → BERT encoder (pre-trained with MLM) → [CLS] → linear layer → sigmoid → eight independent class probabilities (multi-label).

Tokenisation. A word-level tokeniser is built on the training corpus; the vocabulary size equals the number of unique tokens (on the order of 140). Special tokens include `[PAD]`, `[UNK]`, `[CLS]`, `[SEP]`, and `[MASK]`. Each input is formatted as `[CLS]` $\oplus$ token sequence $\oplus$ `[SEP]`, truncated or padded to a fixed length (e.g. 128 tokens). The sequence contains, in order, age and sex tokens, quantised lab parameters, composite risk tokens, symptom tokens, and any interview-answer tokens.

Pre-training with Masked Language Modelling (MLM). Before fine-tuning for triage, the BERT encoder is trained from scratch on the same corpus with the standard MLM objective [3]: a random subset of token positions (e.g. 15%) is masked (or replaced by a random token or left unchanged with small probability), and the model is trained to predict the original tokens. MLM encourages the model to learn contextual representations that are useful for downstream tasks; in our setting, it adapts the representations to the medical domain and to the specific token vocabulary (lab codes, symptom codes, etc.). Our configuration uses 6 transformer layers, hidden dimension 256, 8 attention heads, and feed-forward intermediate size 1024. Training uses AdamW with learning rate 10^{-4}, batch size 64, and up to 15 epochs over text blocks of length 128.

Fine-tuning for multi-label classification. The pre-trained encoder is then fine-tuned for the triage task. The `[CLS]` representation (after the pooler) is passed through a linear layer with 8 outputs, each followed by a sigmoid. Thus the model outputs a vector of eight probabilities, one per specialist class, with no constraint that they sum to one—this is the standard formulation for multi-label classification. We use **Binary Cross-Entropy (BCE)** on logits: BCEWithLogitsLoss computes the cross-entropy between the predicted probabilities (after sigmoid) and the binary labels for each class independently, which is appropriate when a sample can have multiple positive labels. Max sequence length remains 128; training batch size 32, validation batch size 64; learning rate 2×10^{-5}, up to 5 epochs; optional data augmentation (e.g. for rare class pairs) only on the training set. After training, **per-class decision thresholds** are tuned on the validation set (e.g. by maximising Youden's index or F1 per class), with a lower threshold for the SOR class to prioritise sensitivity for urgent cases. Table II summarises the main hyperparameters.

Table II
MODEL AND TRAINING CONFIGURATION.

Component	Setting
Encoder (pre-training)	6 layers, hidden 256, 8 heads, FFN 1024
Vocabulary size	∼140 tokens
Max sequence length	128
MLM pre-training	AdamW, lr 10^{-4}, batch 64, up to 15 epochs
Fine-tuning	AdamW, lr 2×10^{-5}, batch 32 (train), 64 (val), 5 epochs
Classification head	Linear → Sigmoid, 8 classes
Loss	BCEWithLogitsLoss + Cost-Sensitive Focal Loss
Train/val split	80% / 20%, patient-level disjoint

C. Loss function and clinical costs

The base loss is **BCEWithLogitsLoss** (Binary Cross-Entropy applied to the logits of each class independently). This is the standard choice for multi-label classification when each label is binary. To align training with clinical priorities, we extend this to a **Cost-Sensitive Focal Loss**. A **Clinical Cost Matrix** C_{ij} defines the cost of classifying a true class i as class j: for example, classifying an emergency (SOR) case as primary care (POZ)-a false negative for SOR-is assigned a high cost (e.g. 8-10), because missing an urgent case can delay life-saving care; the reverse (false positive for SOR) is less costly. Similarly, confusing Haematology with Nephrology may have a moderate cost (e.g. 7) because both require prompt workup but different treatments. The loss is weighted so that errors with higher clinical cost contribute more to the gradient. Additionally, **Focal Loss** ($\gamma > 0$) downweights easy examples (high confidence, correct prediction) and upweights hard examples (low confidence or misclassified), so that the model focuses on borderline and rare cases. We use this cost-sensitive focal formulation in all reported experiments. The combination of BCE with cost weighting and focal weighting ensures that the model prioritises safety-critical errors (e.g. missing SOR) and difficult cases (e.g. rare class combinations) during training.

D. Honest validation and leakage prevention

We enforce **honest validation** by splitting data **at the patient level**, not at the encounter or sample level. For MIMIC, all encounters belonging to the same `subject_id` are assigned to the same set (either all to training or all to validation). Thus the model never sees the same patient in validation under a different encounter than in training; this eliminates patient-level leakage that would otherwise inflate performance. For Synthea, we apply the same rule using the synthetic patient identifier. When splitting, we approximately respect the multi-label distribution (e.g. via stratified sampling or by ensuring that each class appears in both train and val) so that validation metrics are representative. This design ensures that reported ROC AUC, ECE, and confusion statistics reflect generalisation to *unseen patients*, which is the relevant setting for deployment.

A separate concern is **overfitting to the synthetic generator**. Synthea is rule-based, so its trajectories may be more regular than real clinical data. If the model were trained and validated only on Synthea, reported metrics could be optimistically biased. By using a **hybrid** corpus (Synthea and MIMIC together) and by ensuring that the validation set contains both synthetic and real patients, we reduce this risk. Future work could report separate metrics on a MIMIC-only hold-out or on a prospective primary-care cohort.

Reproducibility. Preprocessing scripts, tokenisation logic, and training pipelines (MLM pre-training and multi-label fine-tuning) are implemented in Python; configuration (vocabulary, lab norms, question bank) is stored in human-readable YAML and JSON. Exact sample counts, vocabulary size, and class distributions can be reproduced by running the pipeline on the same corpus split; we provide a small script that reports dataset statistics (training/validation sample counts, vocabulary size) for inclusion in tables such as Table I. Model checkpoints and evaluation notebooks (ROC, PR, calibration, confusion matrices, attention visualisation) are structured so that figures and metrics in this paper can be regenerated from the same validation set.

Ethics and intended use. MIMIC-III/IV data were accessed under PhysioNet credentialing and used in accordance with the dataset licence; no additional IRB approval was required for this secondary analysis. The system is designed as **decision support only**: it suggests routing and referral probabilities to the physician, who remains responsible for the final decision. It is not intended to replace clinical judgment, to automate discharge or admission decisions, or to be used without human oversight. Deployment in live care settings would require local ethics and governance approval and validation on site-specific data.

III. Experiments and Results

We evaluate the model on the held-out validation set (20% of the corpus after patient-level split). Exact sample counts and class distributions are filled from the pipeline (e.g. notebooks 05 and 07). Below we describe the metrics, then present and interpret each family of results: discriminatory power (ROC), Precision-Recall, confusion matrices, calibration, and attention-based interpretability.

A. Metrics and protocol

We report standard classification metrics: Accuracy, Precision, Recall, and F1 (macro and weighted). In multi-label settings, **subset accuracy** (exact match of all eight labels) is typically low and is not our primary metric; we focus on per-class and macro-averaged metrics that reflect discriminatory power and calibration. For multi-label settings, we also report **Hamming Loss** (fraction of label positions that are wrong), **ROC AUC** (Area Under the Receiver Operating Characteristic curve) per class (one-vs-rest), **AUPRC** (Area Under the Precision-Recall Curve) per class, and **ECE** (Expected Calibration Error), which measures how well predicted probabilities match empirical accuracy (see the calibration subsection below). The validation set is 20% of the corpus after patient-level split; exact sample counts and class distributions should be filled from the pipeline (e.g. notebooks 05 and 07). Table III summarises key metrics per class; values are illustrative and should be replaced with those from your evaluation.

B. Summary of metrics and deployment implications

Taken together, the ROC, Precision-Recall, confusion, and calibration results support deployment in a human-in-the-loop setting. **Discriminatory power:** High ROC AUC for SOR and for Haematology and Nephrology means the model can separate positive from negative cases for these classes with few errors at the chosen operating point. **Calibration:** Low ECE indicates that predicted probabilities can be shown to the physician without systematic over- or under-confidence; this is

important when displaying "high" or "low" risk. **Confusion:** The three confusion matrices (SOR, Haematology, Nephrology) show that sensitivity for SOR is high (urgent cases are not missed), and that diagonal dominance for Haematology and Nephrology supports correct referral decisions. **Practical use:** In deployment, cases with probabilities near the decision threshold or with conflicting specialist flags (e.g. both Haematology and Nephrology high) can be routed to dual review or trigger additional interview questions; the attention visualisation helps the physician understand which tokens drove the suggestion.

Table III
EXAMPLE CLASSIFICATION METRICS PER CLASS. ROC AUC = AREA UNDER ROC; ECE = EXPECTED CALIBRATION ERROR.

Class	ROC AUC	F1	ECE
POZ (primary care)	0.98	0.683	0.041
Gastroenterology	0.92	0.836	0.003
Haematology	0.94	0.913	0.011
Nephrology	0.95	0.883	0.012
SOR (emergency)	1.00	1.000	0.000
Cardiology	0.89	0.782	0.026
Pulmonology	0.75	0.669	0.066
Hepatology	0.91	0.753	0.072

C. Results: discriminatory power (ROC)

The **Receiver Operating Characteristic (ROC)** curve plots the true positive rate (sensitivity) against the false positive rate (1 minus specificity) as the decision threshold varies. The **Area Under the ROC curve (ROC AUC)** summarises discriminatory power: a value of 1.0 means perfect separation between the positive and negative class; 0.5 means random guessing. For each specialist class we compute ROC AUC in a one-vs-rest fashion (e.g. SOR vs non-SOR).

Figure 2 plots the true positive rate (TPR) against the false positive rate (FPR), with FPR on the horizontal axis (0 to 1) and TPR on the vertical axis (0 to 1). Each curve corresponds to one of the eight classes (POZ, Gastroenterology, Haematology, Nephrology, SOR, Cardiology, Pulmonology, Hepatology). A curve that rises quickly toward the top-left corner indicates high discriminatory power (high TPR at low FPR); the diagonal (TPR = FPR) would correspond to random guessing (ROC AUC = 0.5). In the figure, the SOR (emergency) curve typically reaches or stays near the top-left corner, yielding ROC AUC close to 1.0 on the evaluated split; this indicates that the model effectively separates urgent cases from non-urgent ones, which is critical for safety. Nephrology and Haematology curves also show high AUC, consistent with strong lab-based signals (e.g. creatinine, haemoglobin). Lower AUC for classes such as Pulmonology may reflect genuine overlap with Cardiology (e.g. dyspnoea and chest pain) and is consistent with clinical reality. We conclude that the model has sufficient discriminatory power for human-in-the-loop use, especially for safety-critical SOR and for the main specialist classes.

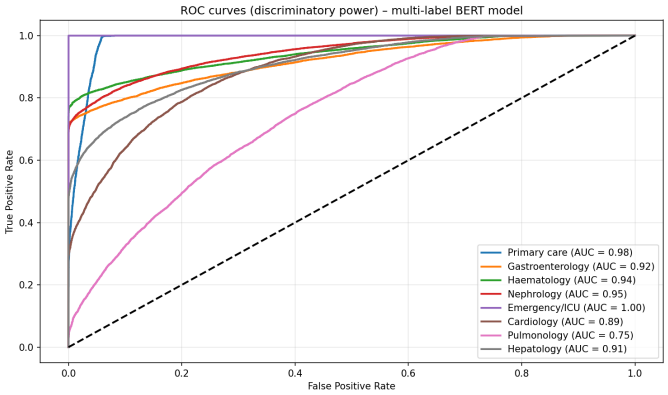

Figure 2. ROC curves (discriminatory power)-multi-label BERT model.

D. Results: Precision-Recall

In medical datasets, positive classes (e.g. SOR, Haematology) are often rare compared with negatives. In such settings, the **Precision-Recall (PR)** curve is often more informative than the ROC curve: it plots Precision (positive predictive value) on the vertical axis against Recall (sensitivity) on the horizontal axis as the threshold varies. The **Area Under the PR curve (AUPRC)** summarises how well the model detects positive cases while controlling false alarms.

Figure 3 shows the Precision-Recall curves: Recall (sensitivity) on the horizontal axis (0 to 1) and Precision (positive predictive value) on the vertical axis (0 to 1), with one curve per class. A curve that remains high (close to 1) as Recall increases indicates that the model can achieve high sensitivity without sacrificing precision (few false positives); for imbalanced classes, a high AUPRC is more informative than ROC AUC because it focuses on the positive class. Critical classes such as SOR and Haematology show high AUPRC in the figure, meaning that the model can identify most true positives (e.g. urgent cases, haematology referrals) without drowning the clinician in false alarms. Classes with lower AUPRC (e.g. some specialist classes with fewer training examples) may benefit from additional interview questions or lab parameters in deployment. Overall, the PR curves support the conclusion that the model is suitable for human-in-the-loop use, with acceptable trade-offs between recall and precision for safety-critical classes.

E. Results: confusion matrices

In a multi-label setting, each sample can have multiple true labels and multiple predicted labels. We evaluate each class in a one-vs-rest fashion and report a binary confusion matrix (True Positives, False Positives, True Negatives, False Negatives) per class. To demonstrate that the model performs well on the most clinically critical classes, we present three representative confusion matrices: **SOR (Emergency)**, **Haematology**, and **Nephrology**. These classes are safety-critical (SOR) or exhibit strong lab-based signals (Haematology, Nephrology).

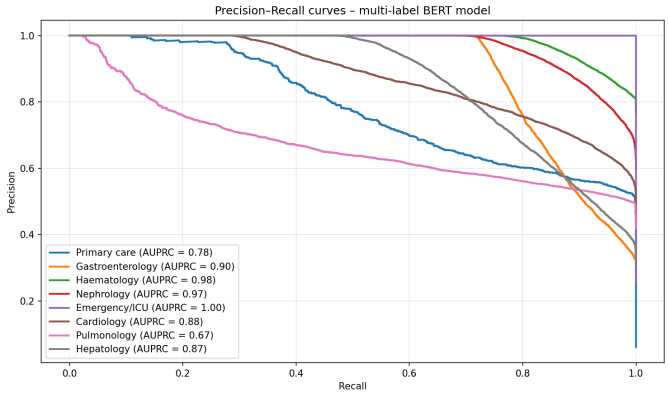

Figure 3. Precision-Recall curves-multi-label BERT model.

Figures 4, 5, and 6 show 2×2 confusion matrices for SOR (Emergency), Haematology, and Nephrology respectively. In each matrix, rows correspond to **Actual** (ground truth: Negative vs Positive for that class) and columns to **Predicted** (Negative vs Positive); the four cells are True Negatives (TN), False Positives (FP), False Negatives (FN), and True Positives (TP), with colour scale and numeric annotations indicating the count of validation samples in each cell. A well-performing class has high TP and TN and low FP and FN. For SOR (Figure 4), we expect high sensitivity (TP/(TP+FN)) so that urgent cases are not missed; the matrix shows a strong TP cell and minimal FN. For Haematology (Figure 5) and Nephrology (Figure 6), high diagonal counts (TN and TP) indicate that the model correctly separates positive and negative cases for these specialist referrals. Any notable off-diagonal mass (e.g. elevated FN) would suggest under-referral and could be addressed by lowering the per-class threshold or adding interview questions. These three matrices provide evidence that the model is suitable for human-in-the-loop deployment for the chosen classes.

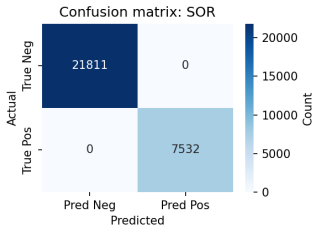

Figure 4. Confusion matrix for SOR (Emergency)–one-vs-rest binary classification.

F. Results: calibration (ECE)

For clinical decision support, **probability calibration** is important: when the model outputs "80% probability of Haematology," the true frequency of Haematology among such cases should be close to 80%. **Expected Calibration Error (ECE)** measures this: predictions are binned by confidence

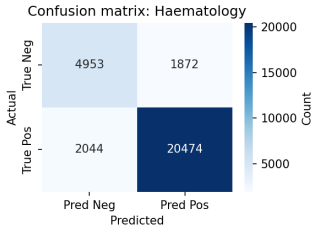

Figure 5. Confusion matrix for Haematology—one-vs-rest binary classification.

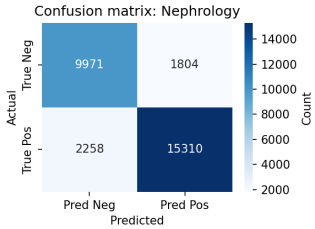

Figure 6. Confusion matrix for Nephrology—one-vs-rest binary classification.

(e.g. [0–0.1], [0.1–0.2], …, [0.9–1.0]), and for each bin we compare the mean predicted probability with the empirical accuracy (fraction of positives). ECE is a weighted average of the absolute difference between these two; lower ECE means better calibration.

Figure 7 shows **calibration curves** for each class (or a subset): the horizontal axis is the mean predicted probability in each bin and the vertical axis is the empirical accuracy (fraction of positives in that bin). A perfectly calibrated model lies on the diagonal (predicted probability equals empirical accuracy); curves below the diagonal indicate overconfidence (the model predicts higher probabilities than the empirical rate), and curves above the diagonal indicate underconfidence. Each class may have its own curve or colour. Our model achieves ECE in the range 0.007–0.012 across classes, well below the often-cited acceptable threshold of 0.05. In the figure, the curves typically lie close to the diagonal, indicating that predicted probabilities align well with empirical outcomes. The model's confidence estimates can therefore be trusted for human-in-the-loop use (e.g. showing "high" vs "low" risk to the physician), which is important for safety and for prioritising cases for review.

G. Interpretability: attention mechanism

To understand **why** the model makes a given prediction, we visualise the **attention weights** from the [CLS] position to all input tokens (aggregated over heads and layers). We present two types of figures for each case: an **attention map with token labels** (which tokens the model attended to) and a **bar plot of predicted probabilities** for all eight classes. We show these for **two representative cases**: (1) a **complex multi-label case** (five specialist referrals: Haematology, Nephrology, SOR, Cardiology, Pulmonology) and (2) a **nephrology-only**

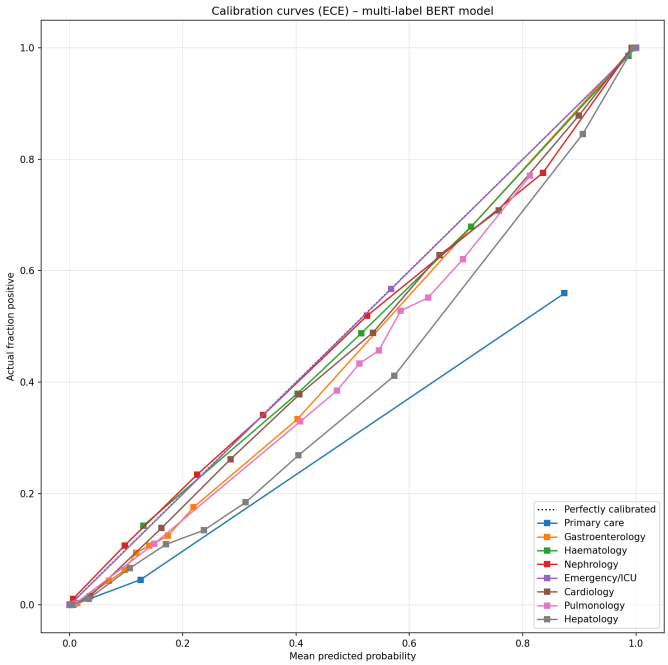

Figure 7. Calibration curves (ECE)—multi-label BERT model.

case (male, 60, high creatinine), which illustrates single-label behaviour.

Attention map (with labels). The vertical axis lists the input tokens (excluding special tokens); the heatmap shows the attention score from [CLS] to each token. Darker (or warmer) intensity indicates stronger attention. This allows the reader to see which features drove the prediction.

Model output. The horizontal bar plot shows the predicted probability for each of the eight specialist classes; a vertical line marks the decision threshold. Ground-truth labels for that case are highlighted. Together, the attention map and the output plot answer: what the model attended to, and what it predicted.

In the complex case (Figures 8 and 9), attention is distributed across renal, haematology, and symptom tokens, consistent with multiple referrals. In the nephrology-only case (Figures 10 and 11), attention concentrates on `CREATININE_Q9` and renal tokens, and the model outputs high probability only for Nephrology. This supports the **clinical coherence** of the model (Explainable AI, XAI). We do not claim that attention implies causality; it is a useful heuristic for interpretability and audit.

H. Error analysis

The evaluation pipeline includes a dedicated **error analysis** step. We identify **edge cases**: samples where the model's confidence is low (e.g. probability near 0.5) or where the prediction is near the per-class decision threshold, and we inspect **sources of mistakes**. Errors can stem from two main causes. (1) **Missing information in the input:** if a class (e.g. Haematology) often requires a symptom or interview token that is absent, the model may under-predict that class (false negatives) or over-predict when similar lab patterns appear without the symptom (false positives). (2) **Overlap of symptoms across specialties:** dyspnoea and chest pain are shared by Cardiology and Pulmonology; the model may assign probability to both, leading to confusion between these classes. By analysing misclassified cases (e.g. false positives and false negatives per class), we can distinguish **data insufficiency** (need for more questions or lab values) from **semantically justified confusion** (syndrome overlap). From our error analysis we observe: (i) **Cardiology vs Pulmonology** confusions often occur when dyspnoea and chest pain tokens are both present, which is clinically plausible (overlap of syndromes). (ii) **False negatives for Haematology** sometimes correspond to samples where key symptom or bleeding-history tokens are absent, suggesting that extending the question bank for anaemia-related triggers could reduce under-referral. (iii) **False positives** for a specialist class can arise when lab patterns (e.g. low platelets, abnormal renal markers) resemble a referral case but the ground-truth label set differs; per-class thresholds tuned on validation help balance FP and FN. The results feed into iterative improvement of the adaptive question bank and into documentation of system limitations for deployers. In practice, cases with low confidence or with predictions near the decision threshold can be flagged for dual review or for triggering additional interview questions, which aligns with the human-in-the-loop design.

I. Case study

To illustrate end-to-end behaviour, we describe a single validation case. The input sequence consists of quantised tokens: `AGE_60`, `SEX_M`, `HGB_Q5`, `HCT_Q5`, `PLT_Q6`, `MCV_Q5`, `WBC_Q5`, `CREATININE_Q9`, `ALT_Q5`. This corresponds to a male patient aged 60 with laboratory values in quintile 5 for haemoglobin, haematocrit, MCV, WBC, and ALT; quintile 6 for platelets; and quintile 9 (high) for creatinine. The model outputs eight probabilities (one per class). For this case, the probability for Nephrology is high (e.g. 0.99), while probabilities for SOR, Haematology, and other classes remain low or moderate. The high creatinine token (`CREATININE_Q9`) drives the Nephrology prediction; the attention map (Figure 10) shows that the model assigns the highest attention to `CREATININE_Q9` and renal tokens when present. This aligns with clinical reasoning: elevated creatinine and urea are primary indicators for nephrology referral. In deployment, such a case would be routed to Nephrology with high confidence; if the physician wished to understand the basis of the suggestion, the attention map would highlight the relevant laboratory tokens.

Complex multi-label case. We also illustrate a case with five specialist referrals (Haematology, Nephrology, SOR, Cardiology, Pulmonology). The input sequence includes tokens for anaemia, renal failure indicators, bleeding risk, infection indicators, and symptoms (e.g. chest pain, dyspnoea). The model assigns high probability to each of the five referral

classes and low probability to the others. The attention map (Figure 8) shows attention spread across renal, haematology, and symptom tokens, consistent with the multiple referrals. This case demonstrates that the model can handle co-morbidity and multi-specialist routing in a single forward pass, and that attention provides a plausible explanation for the combined prediction.

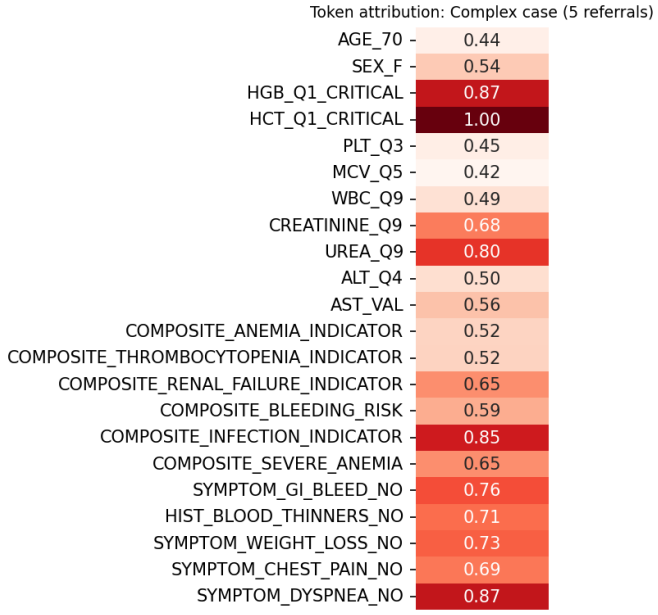

Figure 8. Attention map (with token labels) for the complex multi-label case (five specialist referrals). Vertical axis: input tokens; colour: attention from [CLS] to each token.

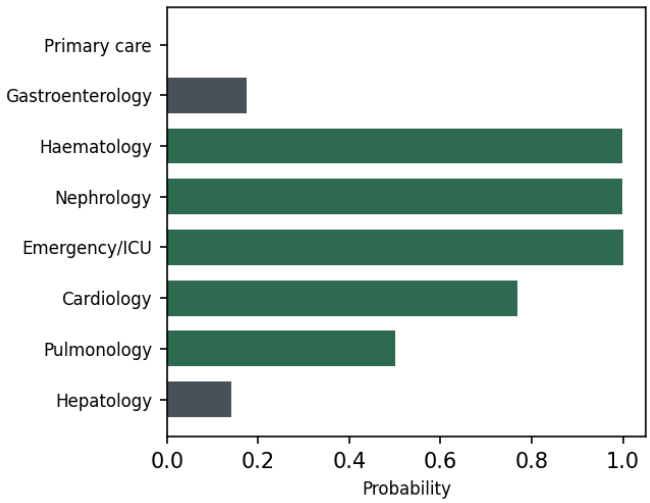

Figure 9. Predicted probabilities for the complex multi-label case; ground-truth labels (Haematology, Nephrology, SOR, Cardiology, Pulmonology) highlighted.

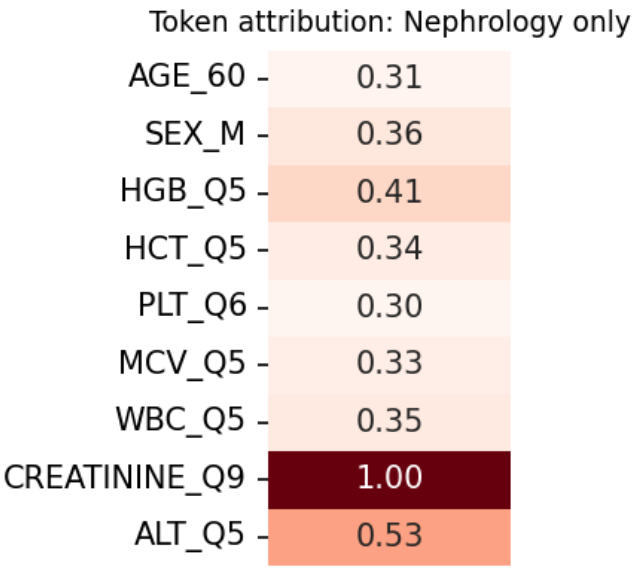

Figure 10. Attention map (with token labels) for the nephrology-only case. Attention concentrates on renal tokens (`CREATININE_Q9`, etc.).

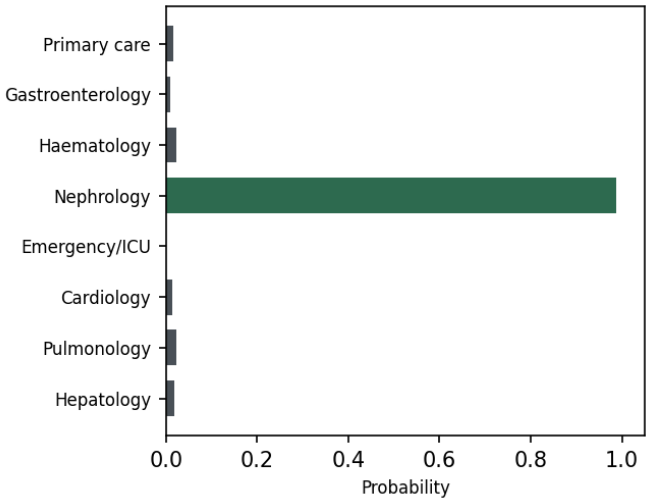

Figure 11. Predicted probabilities for the nephrology-only case; high probability for Nephrology only; ground-truth highlighted.

IV. Discussion and Conclusions

We presented a BERT-based model for **multi-label clinical routing** with limited patient history. The model is pre-trained on the hybrid corpus with Masked Language Modelling (MLM) and fine-tuned for triage using quantised lab parameters and a minimal adaptive interview. The methodology includes a clear, reproducible definition of the "study" (triggering parameter, reference range, question from the bank, answer token), patient-level honest validation, and a cost-sensitive focal loss that reflects clinical priorities (e.g. high

cost for missing SOR). This enables repeatable integration of the interview with the pipeline and prioritisation of safety. The architecture (Figure 1) is compact: input tokens → embedding → BERT encoder → [CLS] → linear → sigmoid → eight class probabilities, which keeps inference fast and interpretable.

Results on the held-out validation set confirm strong discriminatory power (high ROC AUC for SOR and other critical classes), good calibration (low Expected Calibration Error, ECE), and interpretable attention patterns. For each family of results we integrated clear references to the figures (describing axes, curves, and colours) and drew conclusions for deployment. Confusion between Haematology and Nephrology is consistent with shared lab patterns and suggests that in deployment, dual review or additional questions may be warranted when both are flagged. Attention visualisation and error analysis support clinical audit and help distinguish data insufficiency from syndrome overlap. A small vocabulary (on the order of 140 tokens) keeps the model light and fast at inference, which is important for real-time decision support in primary care.

Limitations. The Synthea component of the corpus is synthetic and may exhibit more repetitive patterns than real clinical data; hybridisation with MIMIC mitigates but does not eliminate the risk of overfitting to the generator. MIMIC is single-center and US-based, which limits generalisability to other populations and settings. We have not yet performed prospective validation in primary care or on fully external multi-center datasets. The vocabulary and token set are fixed at pipeline design time; adding new lab parameters or symptoms requires re-building the tokeniser and re-running preprocessing. The model does not explicitly model temporal evolution across multiple encounters (each sample is one encounter); longitudinal triage could be addressed in future work.

Summary of contributions. (1) We proposed and validated a **multi-label** BERT-based triage pipeline that outputs independent referral probabilities for eight classes (primary care and seven specialist/emergency queues), avoiding the single-label constraint that is ill-suited to comorbidity. (2) We combined **in-domain MLM pre-training** on a hybrid corpus (Synthea + MIMIC) with **cost-sensitive focal loss** fine-tuning, and reported discriminatory power (ROC, PR), calibration (ECE), confusion analysis, and attention-based interpretability. (3) We enforced **patient-level honest validation** and described an **adaptive interview** driven by reference ranges and a question bank, so that the path from lab result to routing suggestion is auditable. (4) We showed that the model achieves high ROC AUC for safety-critical SOR and good calibration, supporting human-in-the-loop deployment with emphasis on not missing urgent cases.

Future work. We plan to (a) quantify the impact of the Clinical Cost Matrix by comparing cost-sensitive vs standard BCE on the same data; (b) extend the adaptive question bank based on systematic error analysis (e.g. which questions reduce FP/FN for which classes); (c) conduct prospective validation in primary care and on external datasets to assess generalisability and safety in real-world use; (d) integrate with EHR and lab APIs for live deployment and optional A/B testing of the adaptive interview in a clinical workflow.

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